

## Lecture 7:

Recap: Introduced  $\mathbb{R}$ , as the Cauchy completion of  $\mathbb{Q}$ .

Leftovers: Show that our construction of  $\sqrt{2}$  works.

Recall: We constructed intervals  $I_n = [a_n, b_n]$  s.t.

$$a_n^2 < 2 < b_n^2$$

- $I_{n+1} \subset I_n \quad \forall n.$
- $\text{diam } I_n \leq \frac{1}{2^n}.$
- Let  $x_n \in I_n$  then the sequence  $(x_n)_{n \geq 0}$  is Cauchy. ( $\forall n, m \geq N \quad |x_n - x_m| \leq \frac{1}{2^n}.$ )

Let  $x \in \mathbb{R}$  be the limit of this sequence.

Claim: To show that  $x^2 = 2$ .

let  $f: \mathbb{R} \rightarrow \mathbb{R}$   
 $x \mapsto x^2$

Observe:  $2 \in f(I_n) \quad \forall n$ .

- $f(I_{n+1}) \subset f(I_n)$
- $\text{diam}(f(I_n))$  is ?

let  $y_1, y_2 \in f(I_n)$  compute

$$\begin{aligned} |y_1 - y_2| &= |f(x_1) - f(x_2)| = |x_1^2 - x_2^2| \\ &= |(x_1 - x_2)(x_1 + x_2)| \leq \frac{3}{2^n} \end{aligned}$$

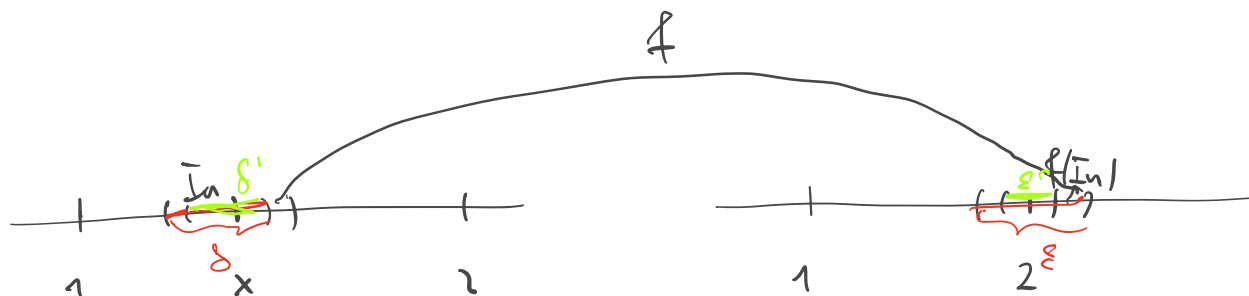
Finally. (Exc)  $x \in \bigcap_{n \geq 0} I_n \Rightarrow f(x) \in \bigcap_{n \geq 0} f(I_n)$

$$\Rightarrow |f(x) - 2| \leq \frac{3}{2^n} \quad \forall n.$$

$$\Rightarrow f(x) = 2.$$

□

Which properties of the function  $f$  did we use in the proof?



Defn let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a function,  $x \in \mathbb{R}$ .

•  $f$  is continuous at  $x$  if

$$\forall \epsilon > 0 \exists \delta > 0 \text{ st. } |x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon.$$

•  $f$  is continuous if it is at all  $x \in \mathbb{R}$ .

Ex. • Constant functions ( $f(x) - f(y) = 0$ ).

•  $f(x) = cx$ . (condition is automatic).

•  $f(x) = x^2$ .

Lemma. If  $f, g: \mathbb{R} \rightarrow \mathbb{R}$  are cont. at  $x$ . Then  
so is their sum and their product.

2nd Observation: we had  $x_n \rightarrow x \quad n \rightarrow \infty$

We also showed that  $f(x_n) \rightarrow \underset{\substack{\uparrow \\ f(x)}}{2} \quad n \rightarrow \infty$

This is a general phenomenon!

Lemma. Let  $f$  be cont. at  $x$ . Then for every  
sequence  $(x_n)_n$  with limit  $x$ , the seq.

$(f(x_n))_n$  converges to  $f(x)$ .

Pf. Pick  $\varepsilon > 0$ .

We want to find  $N$  st.  $\forall n \geq N \quad |f(x_n) - f(x)| < \varepsilon$

Apply continuity of  $f$  gives us

$\delta > 0$  st.  $|x - y| < \delta \Rightarrow |f(y) - f(x)| < \varepsilon$

Applying convergence of  $x_n \rightarrow x$  w/  $\varepsilon = \delta$ .

gives  $N \in \mathbb{N}$  st.  $\forall n \geq N : |x - x_n| < \delta$ .

Combining all of the given  $N \in \mathbb{N} \quad \forall n \geq N$   
 $|f(x_n) - f(x)| < \varepsilon$ .  $\square$

Lemma The converse is also true!

In other words,  $f: \mathbb{R} \rightarrow \mathbb{R}$  is cont. at  $x$

if  $\forall (x: \mathbb{N} \rightarrow \mathbb{R})$  with  $\lim_{n \rightarrow \infty} x_n = x$  we have  
 $\lim_{n \rightarrow \infty} f(x_n) = f(x)$ .

Pf. Assume  $f$  is reg. cont.

Assume by contradiction  $f$  is not  $\varepsilon$ - $\delta$ -cont.

This  $\exists \varepsilon > 0 \forall \delta > 0 \exists y \mid x - y \mid < \delta \wedge \mid f(x) - f(y) \mid \geq \varepsilon$

Want to find a sequence  $x_n$  s.t.

$$1) \quad x_n \rightarrow x \text{ as } n \rightarrow \infty$$

$$2) \quad f(x_n) \not\rightarrow f(x) \text{ as } n \rightarrow \infty.$$

Namely. Fix  $\varepsilon > 0$  from our assumption.

Pick  $x_n$  s.t.  $\mid x - x_n \mid < \frac{1}{n+1}$  and  $\mid f(x) - f(x_n) \mid \geq \varepsilon$

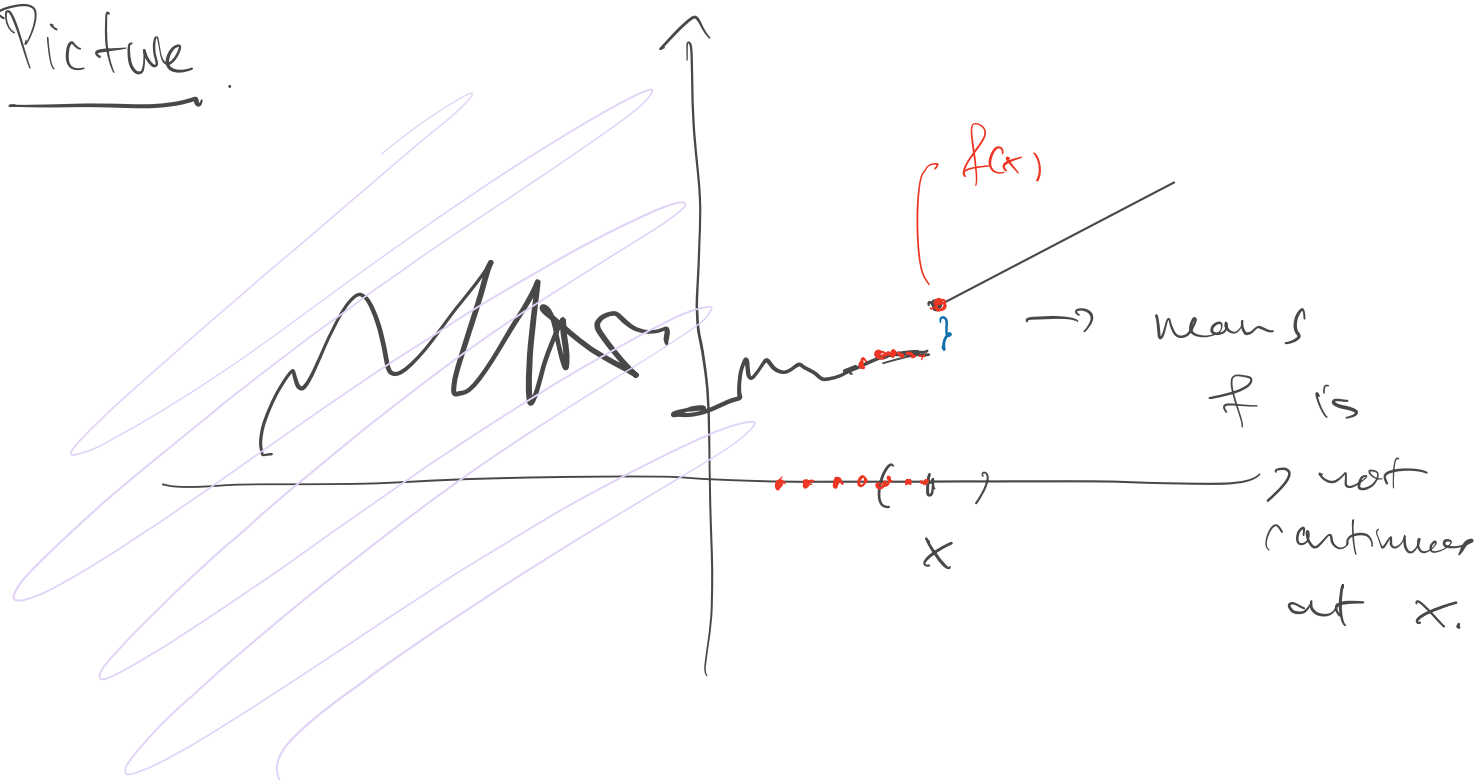
$$\downarrow$$
$$x \rightarrow x_n \text{ as } n \rightarrow \infty$$

$$\downarrow$$
$$f(x_n) \not\rightarrow f(x).$$

Therefore we have found a  
counter. to seq. cont.



Picture



## § Supremum & Infimum

Let  $S \subset \mathbb{R}$  be set.

Defn - We say  $a \in \mathbb{R}$  is an upper/lower bound for  $S$  if  $\forall s \in S \quad a \geq s \quad / \quad a \leq s$ .

- We say  $a \in \mathbb{R}$  is ~~the~~ supremum/infimum of  $S$  if
  - $a$  is upper bound (lower)
  - $\forall b$ , <sup>lower</sup> upper bound of  $S$  :  $a \leq b$ .  $b \leq a$ .

Notation:  $\sup S$  and  $\inf S$ .

Prop Let  $S \subset \mathbb{R}$  be a nonempty set w/ upper bound  $b$ .

Then  $S$  has a supremum.

Pf. Let  $s \in S$ ,  $b$  an upper bound.

Goal. Construct Cauchy seq. st. the limit is the sup. of  $S$ .

We construct intervals  $I_n = [s_n, u_n]$  s.t.

- $I_{n+1} \subset I_n$
- $\text{diam}(I_n) \rightarrow 0$  as  $n \rightarrow \infty$
- $s_n \in S$  and  $u_n$  is upper bound for  $S$ .

$I_n = [s_n, u_n]$   
 $\frac{s_n + u_n}{2}$  is an upper bound.  
if not.  
 $s_n$  any el. of  $S$  which is larger than midpoint.  
Assume for now that we have such  $I_n$ 's.

Then define  $x = \lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} u_n$ .

( $|s_n - u_n| \rightarrow 0$ )

Want to show  $x$  is the supremum of  $S$ .

1) Need to know  $x$  is an upper bound.

Trick:  $u_n \geq s \quad \forall s \in S$

then  $x = \lim_{n \rightarrow \infty} u_n \geq s \quad \forall s \in S$

2)  $\forall b'$ , upper bound of  $S$  we need

$x \leq b'$ .  
 $\lim_{n \rightarrow \infty} s_n$

we know

$s_n \leq b'$

because all  $s_n \in S$ .

$\Rightarrow$  By the same argument  $\square$

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Application

Lemma let  $x_n$  be a monotonically increasing sequence and  $(x_n)_n$  is bounded

$$\text{then } \lim_{n \rightarrow \infty} x_n = \underbrace{\sup \{x_n \mid n \in \mathbb{N}\}}_{=: x}.$$

Proof. Assume by contradiction that  
that  $x_n \not\rightarrow x$ .

$$\exists \varepsilon > 0 \quad \forall N \quad \underbrace{\exists n \geq N} \quad |x - x_n| \geq \varepsilon.$$

Observation  $|x - x_n| \geq \varepsilon \quad \forall n.$

Since  $x_n$  is monotonically increasing  
 $\underbrace{x - x_n}_{\geq 0}$  is monotonically decreasing.

$$\text{For } n \in \mathbb{N} \quad \underbrace{\exists n' \geq n \text{ s.t.}} \quad x - x_{n'} \geq \varepsilon.$$

$$\underbrace{x - x_n}$$

Consider  $x' = x - \frac{\epsilon}{2}$ .

Claim. 1)  $x'$  is an upper bound for  $\{x_n / n \in \mathbb{N}\}$ .

2)  $x' < x$ . ✓

1)

$$\begin{array}{c} x_n < x' < x \\ \underbrace{\hspace{1.5cm}}_{\epsilon/2} \quad \underbrace{\hspace{1.5cm}}_{\epsilon/2} \\ \hspace{1.5cm} \geq \epsilon \end{array}$$

□

Thus  $\exists x'$  contradicts that  $x = \sup\{x_n / n\}$ .

↙

Bonus. Last time we had real numbers as  
a "abstract" completion of  $\mathbb{Q}$ .

Today. Decimal expansions revisited.

In school:  $x \in \mathbb{R}$  then  $x = 1, 12 49 \dots$

Formally: In math terms:

The decimal points are a sequence  $d: \mathbb{N} \rightarrow \text{Fin } 10$

the dots mean the following

$$x = 1 + 1 \cdot 10^{-1} + 2 \cdot 10^{-2} \dots$$

$$= 1 + \sum_{n=0}^{\infty} d_n \cdot 10^{-(n+1)}$$

$$= 1 + \lim_{N \rightarrow \infty} \left( \underbrace{\sum_{n=0}^N d_n 10^{-(n+1)}}_{S_N} \right).$$

Implicitly. All sequences of this form  
converge.

But how to prove this?

1)  $s_n \leq s_{n+1}$  ( $(s_n)_n$  is monotonically increasing)

2)  $s_n$ 's are bounded!

Claim.  $s_n \leq 1 \quad \forall n$ .

$$s_n \leq \sum_{k=0}^n 9 \cdot 10^{-(k+1)} < 1.$$

$$= 9 \cdot \sum_{k=0}^n 10^{-(k+1)}$$

↖ geometric series.

Upshot. Decimal expansion makes sense formally (if converge).

Questions. Is actually every  $x \in \mathbb{R}$   
of this form!